

Hierarchical Forecasting for University Enrollment and Student Performance

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- 1 **Context** — motivation and research questions
- 2 **Data** — exploration and evaluation setup
- 3 **Methodology** — features, models, reconciliation
- 4 **Results** — forecasting, classification, ablation
- 5 **Conclusions** — takeaways, limitations, future work

Why Enrollment Forecasting?

- How many students per course next semester?
- **resource allocation**: rooms, faculty, labs
- Schedule optimisation, curriculum planning

Source: EPSEM-UPC administrative records, 2010–2021.

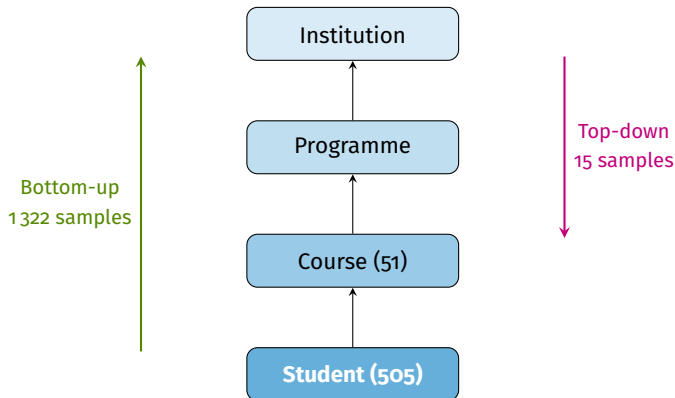
Property	Value
Students	505
Courses	51
Records	11 928
Years	2010–2021
Terms	23
Pass rate	≈70 %

EPSEM-UPC (Manresa)

RQ: How does incorporating individual student academic trajectories into sequential models affect per-course enrolment forecasting accuracy compared to aggregate baselines?

- SRQ1a** How does the choice of student-level architecture (LSTM, GRU, BiLSTM, boosted trees, MLP) affect performance?
- SRQ1b** How sensitive is the best architecture to hyperparameter choices in the small-data regime?
- SRQ2** How does the number of training samples at different aggregation levels affect Micro vs. Macro performance?
- SRQ3** What data-quality obstacles appear when building multi-institutional datasets?

Hierarchical Forecasting

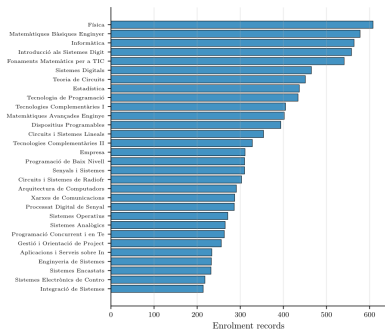


Bottom-up: predict each student's enrolment, then **sum up** to get course-level counts.

Top-down: forecast aggregate counts at institution level, then **distribute** across courses.

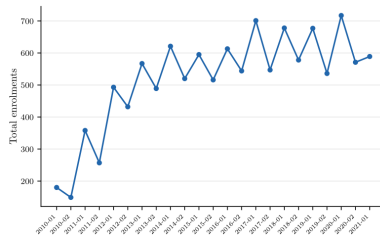
Hierarchical time-series framework adapted from Hyndman et al. (2011).

Enrollment Patterns



Top-30 courses by enrolment count

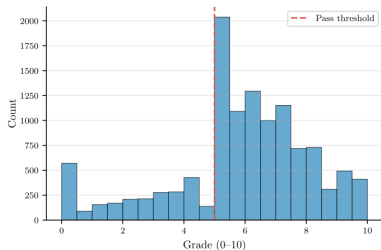
Source: EPSEM-UPC administrative records, 2010–2021.



Total enrolments per academic term

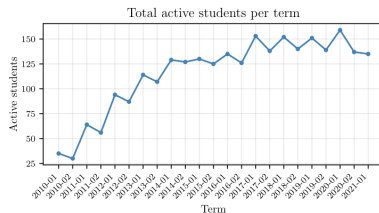
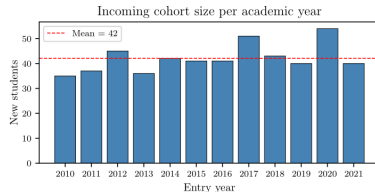


Grade Distribution



Bimodal: peaks near 0 (fail) and 5-7 (pass)

Source: EPSEM-UPC administrative records, 2010-2021.



Data Sources: Multi-University Effort

University	Data	Status
EPSEM-UPC (Manresa, Spain)	505 students, 51 courses, 11 928 records (2010–2021)	Clean and usable Used for all experiments
RTU (Riga, Latvia)	Moodle logs + enrolment. 2 741 codes for ~200 courses	Unusable Platform migrations + No stable identifiers
Vilnius Tech (Lithuania)	Moodle logs + partial enrolment records	Unusable No stable identifiers

Obstacles: platform migrations broke data links, course codes were neither unique nor stable across years; exports lacked documentation.



Mean Absolute Error (MAE): Why This Metric?

Three common regression metrics:

$$\text{MAE} = \frac{1}{n} \sum_i |y_i - \hat{y}_i|$$

$$\text{MSE} = \frac{1}{n} \sum_i (y_i - \hat{y}_i)^2$$

$$\text{RMSE} = \sqrt{\text{MSE}}$$

Example (3 courses):

	Actual	Predicted	Error
Course A	11	10	1
Course B	9	8	1
Course C	20	12	8
MAE		3.33	
MSE		22.0	
RMSE		4.69	

Why MAE for enrollment forecasting?

- Same unit as the target: **students per course**
- Robust to outliers: one high-enrollment course does not dominate
- MSE/RMSE amplify large errors quadratically

Forecasting metric:

$$\text{MAE} = \frac{1}{|C|} \sum_{k=1}^{|C|} |n_k - \hat{n}_k|$$

Averaged over all 51 courses.

Units: students per course.

Classification metric:

Per-course F1-score (tree-based models).

All models are evaluated on the **same 7 test terms** (2018–2021) to ensure comparability.

Train / Test split (chronological):

	Train	Test
Micro	1 322	836
Macro	15	7
Period	2010–17 / 18–21	

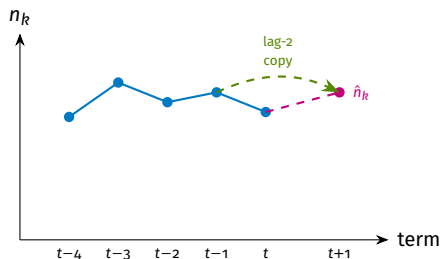
Split at academic year 2018.

No confidence intervals: sequential structure prevents reshuffling.

Naïve Baseline: Lag-2

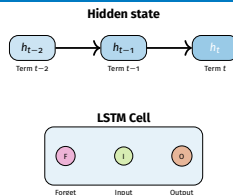
Predicts next semester's enrolment by copying the count of the previous semester (lag-2).

- No parameters, no training
- Very effective!



$$\hat{n}_k^{t+1} = n_k^{t-1} \text{ (same-parity term)}$$

Recurrent Neural Networks and LSTM



LSTM reads a student's term-by-term trajectory and retains relevant patterns across semesters.

Recurrent Neural Networks (RNN) process sequential data step by step, maintaining a hidden state that carries information across time steps.

Problem: standard RNNs lose information over long sequences (vanishing gradients).

Long Short-Term Memory (LSTM) adds a *memory cell* controlled by three gates:

- **Forget:** what to discard from memory
- **Input:** what new information to store
- **Output:** what to pass to the next step

Hochreiter & Schmidhuber (1997). Vanishing gradients: Bengio et al. (1994).



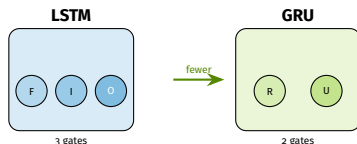
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Gated Recurrent Unit (GRU): Simplified Gating

A **GRU** cell uses two gates (reset and update) instead of the LSTM's three (forget, input, output), reducing parameter count while retaining the ability to model sequential dependencies.

- Fewer parameters → less overfitting on small datasets
- Tested in the ablation study alongside LSTM and Bidirectional LSTM (BiLSTM)

GRU: Cho et al. (2014). BiLSTM: Graves & Schmidhuber (2005).



F=Forget, I=Input, O=Output

R=Reset, U=Update

Classification approach:

- Decision Tree, Random Forest
- XGBoost, LightGBM (gradient boosting)
- Binary prediction per course
→ sum for counts

LightGBM as forecaster:

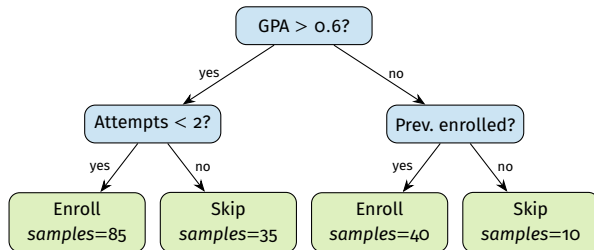
- Flattened 462-dim input
- Also used for direct count regression

XGBoost: Chen & Guestrin (2016). LightGBM: Ke et al. (2017). Word2Vec: Mikolov et al. (2013).

Course2Vec + MLP (Multi-Layer Perceptron):

- Word2Vec on enrolment sequences
- 16-dim course embeddings
- 574-dim input (462 flat + 112 emb.)

How Tree-Based Models Work



A decision tree splits the data based on feature thresholds and based on Gini and Node impurity.

Decision Tree

single tree,
fully interpretable

Random Forest

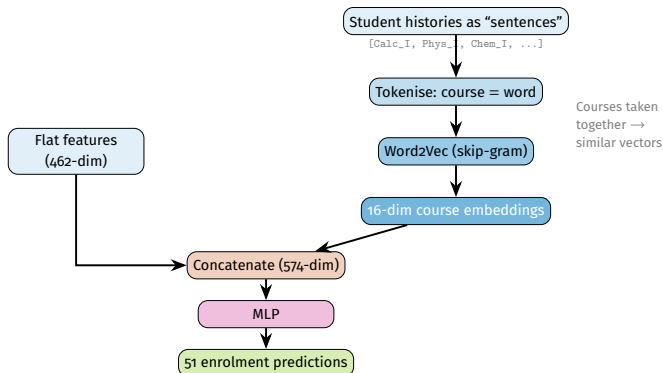
bagging of
decorrelated trees

XGBoost / LightGBM

sequential boosting:
each tree corrects the
residuals (errors) of
the previous ensemble

Course2Vec: Tokenisation Analogy from NLP

Why? The 154-dim vector treats courses independently. Course2Vec learns *relationships between courses* from enrolment patterns, just as Word2Vec learns word relationships from sentences.



Word2Vec: Mikolov et al. (2013).

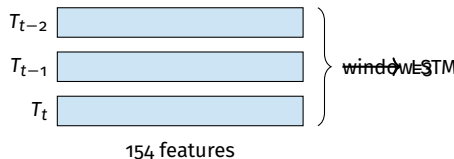
Model	Role	Input	Target	Split
Naïve	Baseline	Lag-2 counts	Count/course	Chrono
Micro LSTM	Sequential	154×3 window	Binary/course	Chrono
Macro LSTM	Sequential	204×3 aggregate	Count/course	Chrono
GRU	Sequential	154×3 window	Binary/course	Chrono
Decision Tree	Tabular	T2 (159 cols)	Binary/course	Random
Random Forest	Tabular	T2 (159 cols)	Binary/course	Random
XGBoost	Boosted	T2 tabular	Binary/course	Random
LightGBM	Boosted	462-dim flat	Binary/course	Both
Course2Vec MLP	Embedding	574-dim flat	Binary/course	Chrono

Chrono = chronological train/test split at 2018. **Random** = 80/20 random split.

All models produce per-course counts by summing predicted probabilities across students.

Feature Vector: 154 Dimensions

$$\mathbf{x}_i = \left[\underbrace{E_i}_{51} \mid \underbrace{G_i}_{51} \mid \underbrace{A_i}_{51} \mid \underbrace{\text{GPA}_i}_1 \right] \in \mathbb{R}^{154}$$



- $E_i \in \{0, 1\}^{51}$: enrolment flags
- $G_i \in [-1, 1]^{51}$: normalised grades (-1 = not taken)
- $A_i \in \{0, \dots, 5\}^{51}$: cumulative attempts
- $\text{GPA}_i \in [0, 1]$: cumulative average

Now that we have seen all models, how do we apply them?

Micro (Student-level)

- 1322 training samples
- Each sample = one student's 3-term trajectory
- Input: 3×154 (feature vector)
- Predict per-student enrolment $\rightarrow$ sum for counts

Macro (Term-level)

- 15 training samples
- Each sample = one term's aggregate statistics
- Input: 204-dim ($4 \text{ stats} \times 51 \text{ courses}$)
- Predict course counts directly

88:1 sample ratio

Micro input: $[E, G, A, \text{GPA}]$ per student per term = $154\text{-dim} \times 3 \text{ terms}$

Macro input: $[\text{mean, std, min, max}]$ of enrolment per course = $4 \times 51 = 204\text{-dim}$

Five ways to combine Micro and Macro forecasts into a single prediction:

Strategy	Formula
Simple average	$\tilde{y}_k = \frac{1}{2} (\hat{y}_k^{\text{mi}} + \hat{y}_k^{\text{ma}})$
Global weighted	$\tilde{y}_k = \alpha \cdot \hat{y}_k^{\text{mi}} + (1-\alpha) \cdot \hat{y}_k^{\text{ma}}$
Per-course weighted	$\tilde{y}_k = \alpha_k \cdot \hat{y}_k^{\text{mi}} + (1-\alpha_k) \cdot \hat{y}_k^{\text{ma}}$
MinT (Minimum Trace) proportional	$\tilde{y}_k = \hat{y}_k^{\text{mi}} + D \cdot \frac{\hat{y}_k^{\text{mi}}}{\sum_j \hat{y}_j^{\text{mi}}}$
Micro-Naïve blend	$\tilde{y}_k = \alpha \cdot \hat{y}_k^{\text{mi}} + (1-\alpha) \cdot \hat{y}_k^{\text{naïve}}$

Key question: does combining **improve** over standalone Micro?

Reconciliation framework: Hyndman et al. (2011); MinT: Wickramasuriya et al. (2019).

Averages the Micro and Macro predictions equally, treating both sources as equally reliable.

$$\tilde{y}_k = \frac{1}{2} (\hat{y}_k^{\text{mi}} + \hat{y}_k^{\text{ma}})$$

Example (2-course hierarchy):

	Micro	Macro	Reconciled	Actual
Course A	10	16	13.0	11
Course B	8	13	10.5	9
MAE	1.0	4.5	1.75	—

Reconciliation: Global Weighted Average

Combines Micro and Macro with a single weight α optimised across all courses via grid search.

$$\tilde{y}_k = \alpha \cdot \hat{y}_k^{\text{mi}} + (1-\alpha) \cdot \hat{y}_k^{\text{ma}}$$

Example ($\alpha = 1.0$, the optimum):

	Micro	Macro	Reconciled	Actual
Course A	10	16	10.0	11
Course B	8	13	8.0	9
MAE	1.0	4.5	1.0	—

Reconciliation: Per-Course Weighted

Learns a separate weight α_k for each course, allowing the blend to vary where Macro happens to be locally useful.

$$\tilde{y}_k = \alpha_k \cdot \hat{y}_k^{\text{mi}} + (1 - \alpha_k) \cdot \hat{y}_k^{\text{ma}}$$

Example ($\alpha_A=1.0$, $\alpha_B=0.8$):

	Micro	Macro	Reconciled	Actual
Course A	10	16	10.0	11
Course B	8	13	9.0	9
MAE	1.0	4.5	0.5	—

Reconciliation: Micro–Naïve Blend

Replaces the unreliable Macro with the Naïve baseline as the second component, blending two individually competitive forecasters.

$$\tilde{y}_k = \alpha \cdot \hat{y}_k^{\text{mi}} + (1-\alpha) \cdot \hat{y}_k^{\text{naïve}}$$

Example ($\alpha = 0.5$):

	Micro	Naïve	Reconciled	Actual
Course A	10	11	10.5	11
Course B	8	9	8.5	9
MAE	1.0	0.0	0.5	—

Reconciliation: MinT (Minimum Trace) Proportional

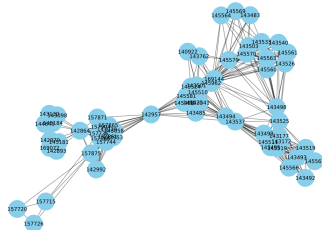
The **discrepancy** D is the gap between the Macro total and the Micro total: if Macro predicts 29 students overall but Micro predicts only 18, then $D = 11$ students are “missing”. MinT redistributes this gap proportionally to each course’s Micro forecast.

$$\tilde{y}_k = \hat{y}_k^{\text{mi}} + D \cdot \frac{\hat{y}_k^{\text{mi}}}{\sum_j \hat{y}_j^{\text{mi}}}, \quad D = \sum_j \hat{y}_j^{\text{ma}} - \sum_j \hat{y}_j^{\text{mi}}$$

Example ($D = 29 - 18 = 11$):

	Micro	Macro total	Reconciled	Actual
Course A	10	29	16.1	11
Course B	8		12.9	9
MAE	1.0	—	4.5	—

Programme Cluster 10 (64 courses)



Course-code redundancy:

- ~8–10 suspected renames
- Same course, different code
- Inflates course count artificially

Multi-university acquisition:

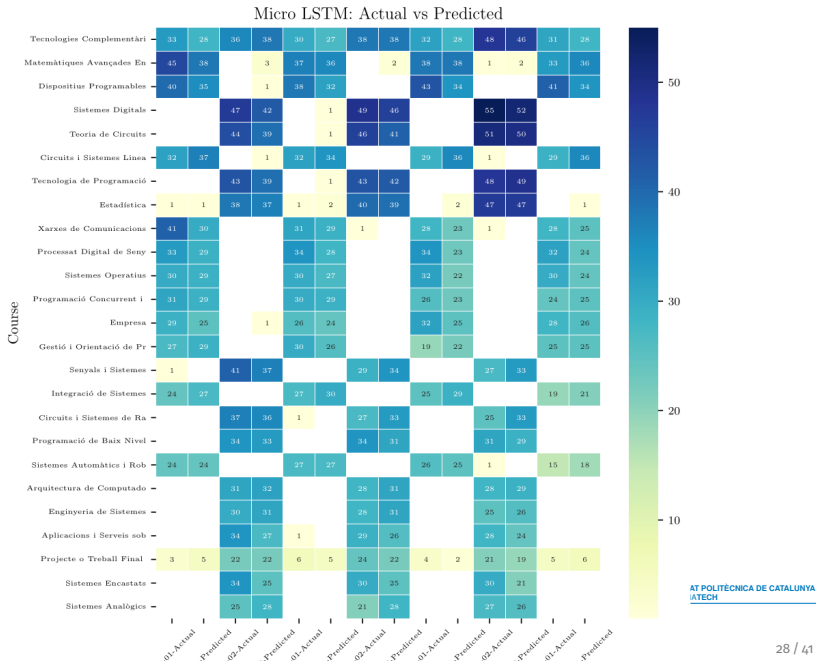
- RTU (Latvia): 2 741 codes for ~200 courses
 - Vilnius Tech: no structured export
- Proposed *.educast.json format

Co-enrolment graph from RTU data.
Nodes = courses, edges = students
enrolled in both. Clusters reveal
programme structure.

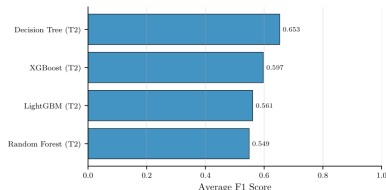
Model	MAE (students/course)
LightGBM (student-level)	1.68
Naïve baseline (lag-2)	1.69
Micro LSTM	1.73
Course2Vec MLP	1.78
Macro LSTM	5.57

Mean Absolute Error on 2018–2021 test set (7 terms), computed over all 51 courses.

Micro LSTM: Forecast Heatmap



Classification Results

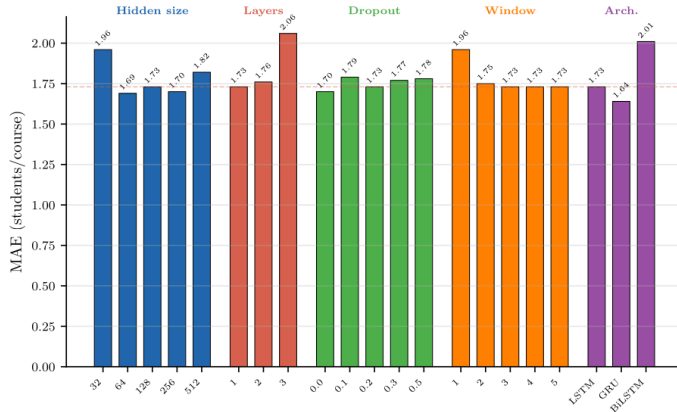


Model	F1
Decision Tree	0.653
XGBoost	0.597
LightGBM	0.595
Random Forest	0.549



Depth-4 tree for **Senyals i Sistemes**:
reveals curricular progression patterns.

Ablation Study

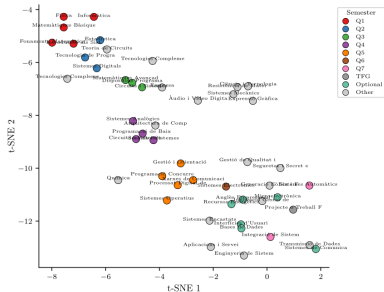


- **Architecture:** LSTM (1.73) > BiLSTM (2.01)
- **Layers:** 1 layer best, more → overfitting

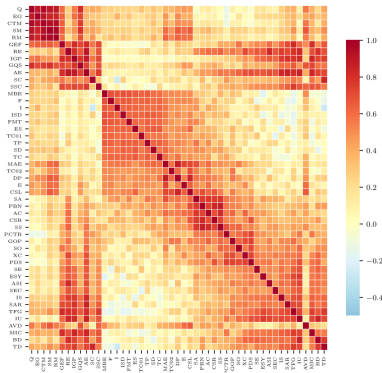
- **Hidden size:** 64-256 similar
- **Window:** 3-5 equivalent



Course Embeddings (Course2Vec)



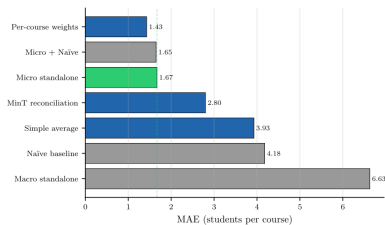
Courses from the same semester
cluster together automatically



Cosine similarity matrix reveals
prerequisite and co-requisite chains

t-SNE (t-distributed Stochastic Neighbor Embedding) compresses the 16-dim vectors into 2D: courses close in the embedding space appear close in the plot. Course2Vec MLP (1.78) improves over plain MLP (1.94).

Reconciliation Results



Method	MAE
Micro standalone	1.73
Weighted ($\alpha=1.0$)	1.73
MinT	2.80
Simple average	3.93
Macro standalone	6.63

Optimal $\alpha = 1.0$: Macro contributes **nothing**.
MinT (Minimum Trace) **injects** Macro noise $\rightarrow$ **+62%** worse than
standalone Micro.

Comparison with State of the Art

Study	Dataset	Model	Finding
Khan & Polyzou (2024)	FIU (Florida): 3 324 students, 334 courses	Recommender repurposing	Rec. methods beat baselines
This thesis	EPSEM (Manresa): 505 students, 51 courses	LSTM / GRU / LightGBM	Micro matches Naïve

- Our dataset is 6.6× smaller than FIU
- Absolute MAE values not directly comparable across institutions
- Student-level approach is viable even at small scale

Khan & Polyzou (2024), *J. Educ. Data Mining*, 16(1).

- 1 **Student-level models are viable:** Micro LSTM achieves MAE 1.73, competitive with Naïve outperforming by a small margin.
- 2 **Macro fails at small scale:** 15 training windows vs. 1322 → MAE 5.57. The 88:1 sample ratio is too extreme.
- 3 **Reconciliation needs good components:** combining a good Micro with a poor Macro makes predictions **worse**.
- 4 **Data quality is the real bottleneck:** course-code redundancy and small cohort limit improvement over Naïve.

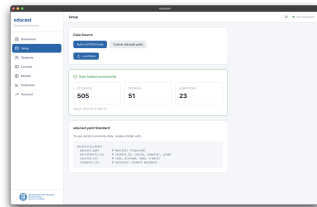
External validity Single institution (EPSEM-UPC). Results may not generalise to larger universities.

Statistical conclusion Small test set (7 terms for Macro). Tree-based models use random split → not directly comparable.

Programme rules No explicit encoding of "prerequisites" or certain rules such as course time schedules, which may affect enrolment patterns.

- 1 **154-D feature representation** encoding enrolment, grades, attempts, and GPA per term.
- 2 **Micro vs. Macro comparison** showing student-level modelling produces 88× more training samples.
- 3 **Reconciliation evaluation:** five strategies tested, showing that reconciliation fails when aggregate forecasts are poor.
- 4 **Data-quality documentation** from multi-university acquisition (RTU, Vilnius Tech).
- 5 **Standardised format** (*.educast.json) for institutional enrolment records.

The Educast Framework



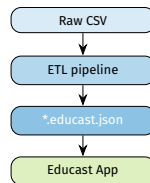
Pydantic schema (*.educast.json):

```
class AttemptedCourse(BaseModel):
    course: Course
    year: Optional[int]
    grade: Optional[float]
    clickstream: List[ClickstreamData]

class Student(BaseModel):
    student_id: str
    history: AcademicHistory

class University(BaseModel):
    name: str
    departments: List[Department]
    students: List[Student]
```

Educast (Electron app)



Load → explore → train → evaluate

Short-term:

- Learnable course embeddings (end-to-end)
- Moodle / LMS (Learning Management System) engagement features such as clickstream data

Medium-term:

- Transformer-based architectures
- Schedule / timetable features
- Institutional enrolment rules

Long-term:

- Multi-institution data harmonisation
- Hierarchical reconciliation at scale
- Test it in larger institutions / datasets

Thank you

Questions?

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